Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/16/2021
PI Name:	Willey, Jeffrey S. Ph.D.		
Project Title:	Exercise Countermeasures for Knee and Hip Joint Degradation during Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	27157-0001	**Congressional District:**	5
Comments:	NOTE: PI formerly at Clemson University when NSBRI Postdoctoral Fellow Feb 2008-Oct 2010 (Ed., 12/18/2014)		
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	10/28/2014	**End Date:**	05/31/2021
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:	ISS Rodent Research-9 NOTE: End date changed to 5/31/2021 per F. Hernandez/ARC and NSSC information (Ed., 5/5/21) NOTE: End date changed to 3/31/2021 per F. Hernandez/ARC (Ed., 4/7/2020) NOTE: End date changed to 3/31/2020 per F. Hernandez/ARC (Ed., 6/23/17)		
Key Personnel Changes/Previous PI:	NOTE September 2021: Dr. Ted Bateman was no longer a CoI during the last year of the project, from September 2020-May 2021.		
COI Name (Institution):			
Grant/Contract No.:	NNX15AB50G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Maintaining musculoskeletal health during long-duration spaceflight is crucial for ensuring both mission success and full skeletal recovery upon returning to weight-bearing. Clinical and preclinical evidence indicates that cartilage degradation in the hip and knee joints occurs with reduced weight-bearing. Less well characterized are the damaging effects of spaceflight-relevant radiation on cartilage, including exposure to solar particle events (SPE). Deterioration of the hip and knee joint during prolonged spaceflight has the potential to reduce an astronaut's performance during a mission, cause arthritis, and negatively impact the astronaut's long-term quality of life (QOL). Our study will test the hypothesis that mouse hip and knee joints exposed to microgravity on the International Space Station (ISS) or from reduced weight bearing via tail-suspended with or without exposure to spaceflight-relevant doses of radiation in Definition Phase studies will exhibit profound tissue degradation. Additionally, this degradation can be recovered using aerobic (running) and resistance (climbing) exercise countermeasures. To study these problems, we will determine the hip and knee joint damage that occurs in mice that will fly in space on the International Space Station for 30 days. This damage will be compared to the hip and knee joint damage in another group of mice kept on Earth that also will not have weight on the hip and knee joints for 30 days, with or without receiving radiation exposure that simulates a solar flare. Damage to the hip and knee joint structures will be determined using imaging techniques, engineering devices to measure tissue strength, stained tissue sections, and identification of the molecules that cause the damage. The ability to walk normally after 30 days of weightlessness will also be determined. Finally, we will determine if treadmill running or climbing can reverse any of the hip and knee joint damage caused by being in the weightless space environment. Our goal is to determine, 1] if hip and knee joint damage occurs in the weightless space environment, and 2] if recovery from this damage is possible with exercise.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	From these studies, we also will gain insights into how arthritis and joint failure develop in both patients that receive radiation therapy for the treatment for cancer, and in patients with limited mobility (cancer patients, wheel-chair bound spinal cord injury patients, or after limb surgery), and how this can be prevented.
Task Progress:	FINAL REPORTING SEPTEMBER 2021 With a return to lab from COVID restrictions, we compiled and wrote a manuscript that incorporated all the finished data from our ISS mission, together with the parallel ground-based tail suspension study (with and without a readaptation period that included exercise interventions or not), and also included data from knee joint degradation that were collected during the final Space Shuttle mission (STS-135). Our final conclusion was that spaceflight and reduced weight-bearing on the ground induce an arthritic response in the knees of mice, which can be reversed with a return to weight-bearing (after hindlimb unloading-HLU) if exercise is performed. This final paper was submitted to Scientific Reports and published in the fall. Moreover, the student who was working on the project as his dissertation research, Andy Kwok, wrote and defended his dissertation in Spring, 2021. [Ed. note: see Bibliography section for noted publications] ANNUAL REPORTING SEPTEMBER 2020 We examined if reduced weight bearing on Earth using a hindlimb unloading (HU) model could cause damage to menisci and cartilage, as during spaceflight. What we found was that in male and female mice, the degree of cartilage and meniscus damage was similar to what was observed after 30 days in space, and occurred at the same location (the region in the knee that supports most of the weight versus gravity). Thus cartilage degradation, and meniscal loss, occurs with reduced weight bearing, and HU is a good analogue for joint damage that occurs during spaceflight. We then identified that both running and climbing exercise can in part mitigate the damage to cartilage and menisci that is caused by reduced weight bearing.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
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